CENTRE FOR INNOVATION & SYSTEMS AI

Raspberry Pi Server Installation & Operations Guide

Production Reference: Pi-4/5 LTS

Target OS: Ubuntu 24.04 LTS (64-bit)
Host IP: • Port: 8080 / 8443

1. Overview and Architecture

This document outlines the complete deployment procedure for running the **Central TV Streaming & Signage Hub Server** directly on a Raspberry Pi host. Once deployed, administrative personnel and operators can access the central control portal from any laptop on the corporate network without installing local client software.

Core Server Responsibilities

- **Web Application Portal:** Serves responsive Admin Console, Executive Dashboard, and Presenter interfaces via HTTP/HTTPS.
- **Device Fleet Registry:** Manages real-time bidirectional WebSocket connections with all connected TV endpoints.
- **Media Library Manager:** Stores and delivers 1080p/4K signage videos with dual-buffer seamless looping.

MediaMTX Live Streaming Engine

- **WebRTC (WHEP/WHIP):** Sub-300ms ultra-low latency real-time screen sharing and camera broadcast.
- **HLS Server:** Low-latency segmented streaming fallback for legacy browser players.
- **RTSP / RTMP Ingest:** Ingestion points for OBS Studio, hardware encoders, and IP security cameras.

2. Hardware & Operating System Prerequisites

Component	Recommended Specification	Verified Target Environment
Hardware Platform	Raspberry Pi 5 (8GB) or Pi 4 Model B (4GB/8GB)	Raspberry Pi 4 / 5 (8GB RAM, aarch64)
Operating System	Ubuntu Server 24.04 LTS Noble Numbat (64-bit)	Ubuntu 24.04.5 LTS (Kernel 6.8.0-1064-raspi)
Storage	Class 10 A2 MicroSD or M.2 NVMe SSD (32GB+)	32GB High-Speed Storage (23GB Free)
Network	Gigabit Ethernet (Static IP Recommended)	Ethernet Interface:
Power Supply	Official 27W USB-C Power Adapter (Pi 5) / 15W (Pi 4)	Stable regulated power line

3. Step-by-Step Installation Procedure

3.1 Initial System Updates and Core Package Installation

Connect to the Raspberry Pi over SSH and install the required runtimes, build tools, and FFmpeg:

```
# Connect to Raspberry Pi
ssh ubuntu@

# Update package repositories and install runtimes
sudo apt update && sudo apt install -y   nodejs npm ffmpeg rsync curl jq build-essential
```

Verify that Node.js (v18+ LTS) and FFmpeg are correctly installed:

```
node -v     # Output: v18.19.1 or higher
npm -v      # Output: 9.2.0 or higher
ffmpeg -version | head -n 1
```

3.2 Application Directory Setup & File Synchronization

Create the designated application root at `/opt/central-tv-streaming` and configure correct ownership:

```
sudo mkdir -p /opt/central-tv-streaming
sudo chown -R ubuntu:ubuntu /opt/central-tv-streaming
```

From your workstation, synchronize the server codebase, assets, and media repository using `rsync`:

```
# Execute from your development workstation
rsync -avz --exclude '.git' --exclude 'yt-dlp' --exclude 'media_4k_backup'   ./central-tv-streaming/ ubuntu@:/opt/central-tv-
```

Install production Node.js dependencies on the Raspberry Pi:

```
cd /opt/central-tv-streaming
npm install --omit=dev
```

> **Included Runtime Modules:** `express` (v5 web framework), `ws` (high-concurrency WebSocket engine), `multer` (chunked media upload handler), and `cookie-parser` (session authentication security).

4. MediaMTX Streaming Engine Setup

MediaMTX is deployed as a standalone native ARM64 binary to avoid container virtualization overhead and ensure hardware-accelerated socket throughput on the Pi:

```
# Download native ARM64 release binary directly on the Pi
curl -sL https://github.com/bluenviron/mediamtx/releases/download/v1.9.0/mediamtx_v1.9.0_linux_arm64v8.tar.gz |   tar -xz -C

chmod +x /opt/central-tv-streaming/mediamtx
```

Ensure the configuration file at `/opt/central-tv-streaming/mediamtx.yml` contains verified settings:

```
# /opt/central-tv-streaming/mediamtx.yml
logLevel: info
logDestinations: [stdout]

api: yes
apiAddress: :9997

rtsp: yes
rtspAddress: :8554
protocols: [udp, multicast, tcp]

rtmp: yes
rtmpAddress: :1935

hls: yes
hlsAddress: :8888
hlsSegmentCount: 7
hlsSegmentDuration: 1s
hlsPartDuration: 200ms

webrtc: yes
webrtcAddress: :8889
webrtcEncryption: no
webrtcAllowOrigin: "*"
webrtcLocalUDPAddress: :8189
webrtcIPsFromInterfaces: yes
webrtcHandshakeTimeout: 10s

paths:
  all:
```

5. Systemd Daemon Configuration & Auto-Boot

To ensure high availability and automatic recovery following power outages or network reboots, two systemd services are created:

5.1 MediaMTX Service: `/etc/systemd/system/mediamtx.service`

```
[Unit]
Description=MediaMTX Streaming Core (RTSP, RTMP, WebRTC, HLS)
After=network.target network-online.target
Wants=network-online.target

[Service]
Type=simple
User=ubuntu
Group=ubuntu
WorkingDirectory=/opt/central-tv-streaming
ExecStart=/opt/central-tv-streaming/mediamtx /opt/central-tv-streaming/mediamtx.yml
Restart=always
RestartSec=3
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
```

5.2 Central TV Server Service: `/etc/systemd/system/central-tv-server.service`

```
[Unit]
Description=CiSAI Central TV Streaming & Signage Hub Server
After=network.target network-online.target mediamtx.service
Wants=network-online.target

[Service]
Type=simple
User=ubuntu
Group=ubuntu
WorkingDirectory=/opt/central-tv-streaming
ExecStart=/usr/bin/node server.js
Restart=always
RestartSec=3
Environment=NODE_ENV=production
Environment=HTTP_PORT=8080
Environment=HTTPS_PORT=8443
Environment=MEDIAMTX_HOST=127.0.0.1
Environment=MEDIAMTX_WEBRTC_PORT=8889
Environment=MEDIAMTX_HLS_PORT=8888
Environment=ADMIN_USER=admin
Environment=ADMIN_PASSWORD=admin123
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
```

5.3 Enabling and Launching Services

```
sudo systemctl daemon-reload
sudo systemctl enable --now mediamtx.service
sudo systemctl enable --now central-tv-server.service
```

6. Verification & Network Access from Laptops

Once started, the central server is accessible over the local network. Open any web browser on your connected laptop and navigate to the following endpoints:

Portal Interface	Network URL	Purpose & Usage
Admin Control Console	`http://:8080/admin.html`	Fleet control, media playlist management, channel switching, TV reboot
Executive Dashboard	`http://:8080/dashboard.html`	Multi-screen live operations monitoring & telemetry
Live Presenter / Screen Cast	`http://:8080/presenter.html`	Real-time screen broadcasting & microphone streaming
TV Player Endpoint	`http://:8080/player.html?name=TV1&site=HQ`	Display screen client player with seamless loop engine
Encrypted HTTPS Portal	`https://:8443`	Secure TLS encrypted management access

Role-Based Access Control (RBAC) Accounts

Role	Username	Password	Permitted Operations
Administrator	`admin`	`admin123`	Full system control, device fleet discovery, TV deletion, alerts, channel setup
Streamer / Operator	`streamer`	`streamer123`	Upload videos to loop, live presenter screen/camera stream, TV sync & override

Credentials can be configured via environment variables `ADMIN_USER`, `ADMIN_PASSWORD`, `STREAMER_USER`, `STREAMER_PASSWORD` in systemd or via `data/users.json`.

6.1 Dual-Interface Network Configuration (Netplan)

Production Raspberry Pi hubs utilize dual network interfaces to segregate internet/egress streaming traffic from the internal TV management VLAN:

Interface	IP / Netmask	Gateway & Metric	DNS Nameservers
eth0 (Built-in Port)	`10.31.93.5/23` (255.255.254.0)	`10.31.93.2` (Metric: 100)	`8.8.8.8`
enx... (USB Adapter)	`10.31.1.190/23` (255.255.254.0)	`10.31.0.1` (Metric: 200)	`10.31.4.1`, `10.31.4.3`

Verified Netplan configuration file (`/etc/netplan/50-cloud-init.yaml`):

```
# /etc/netplan/50-cloud-init.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    eth0:
      dhcp4: false
      addresses:
        - 10.31.93.5/23
      routes:
        - to: default
          via: 10.31.93.2
          metric: 100
      nameservers:
        addresses:
          - 8.8.8.8
    enx207bd28aa480:
      match:
        macaddress: 20:7b:d2:8a:a4:80
      set-name: enx207bd28aa480
      dhcp4: false
      addresses:
        - 10.31.1.190/23
      routes:
        - to: default
          via: 10.31.0.1
          metric: 200
      nameservers:
        addresses:
          - 10.31.4.1
          - 10.31.4.3
```

Apply changes with: `sudo netplan apply`

7. Operations, Maintenance & Troubleshooting

Service Health Check Commands

```
# Check server status
systemctl status central-tv-server.service

# Check streaming engine
systemctl status mediamtx.service

# Inspect live application logs
sudo journalctl -u central-tv-server.service -f
```

System Resource Verification

```
# Check RAM usage (Under 600MB typical)
free -h

# Check disk space
df -h /

# Check CPU temperature
cat /sys/class/thermal/thermal_zone0/temp
```

Port Allocation Reference

Port	Protocol	Service	Description
`8080`	TCP	Node.js Express / WebSocket	HTTP Web Console, Device WebSocket API, Media streaming
`8443`	TCP	Node.js HTTPS	Secure TLS Web Console
`8889`	TCP	MediaMTX WebRTC	WebRTC HTTP Signaling (WHIP/WHEP)
`8189`	UDP	MediaMTX ICE	WebRTC Low-Latency Video Packets
`8888`	TCP	MediaMTX HLS	HTTP Live Streaming segmented video
`8554`	TCP/UDP	MediaMTX RTSP	RTSP video feed ingestion
`1935`	TCP	MediaMTX RTMP	RTMP video ingestion for OBS Studio